II-Probabilistic Approaches Causality

Foundations of Neuro-Symbolic AI

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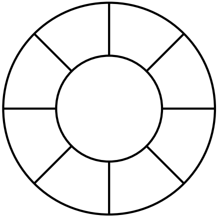
Outline

- Bayesian network hypergraphs
- Intervention
- Counterfactuals

Example: Random walk

Imagine a random walk through circular cells:



- X_k describes at timestep $k \in [d]$ the position
- With probability 0.5 we stay at a cell, and to each neighboring cell with 0.25

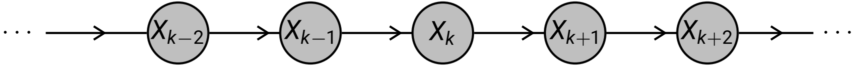
This is a Markov chain with the transition matrix:

$$\mathbb{P}\left[X_k \middle| X_{k-1}\right] = \begin{pmatrix} 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 \\ 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 \\ 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 \end{pmatrix} [X_k, X_{k-1}]$$

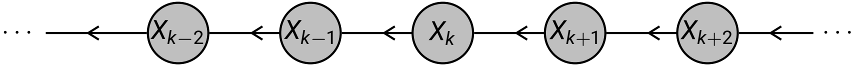
How to distinguish the past from the present?

We have two decomposition schemes as Bayesian networks, both decorated with the same transition matrices:

Parametrization from the past to the future:



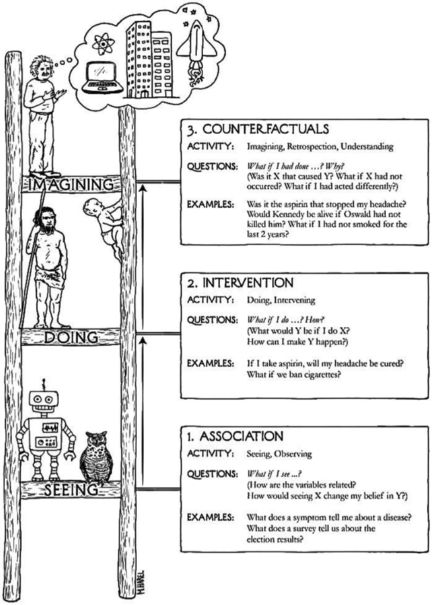
Parametrization from the future to the past:



The BN hypergraph is not unique

We can manipulate the hypergraph, and still find a Bayesian network representation of a distribution.
Causation is not (always) represented in the directionality!
Idea: Distinguish by interventions.

The rungs of causation



→ **Response variables**

→ **Do variables**

→ **Bayesian Networks**

Bayesian networks capture associations

Limitation: Do not model causations.

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Interventions

This appears in important situation:

- Will product sales increase if we lower the price?
- Will I pass the exam tomorrow if I study today?

Do calculus: Modelling interventions

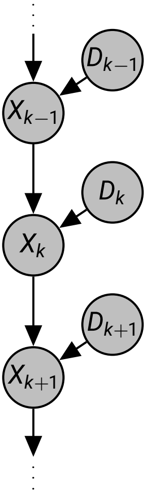
Add further variables D_k representing interventions:
Ignore the state of a variables parents.
This resolves the ambiguity in Bayesian network representation!

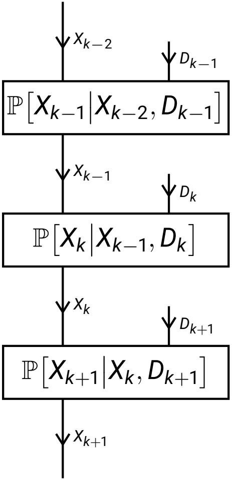
Example: Random walk

Random walk example with $m = 3$:

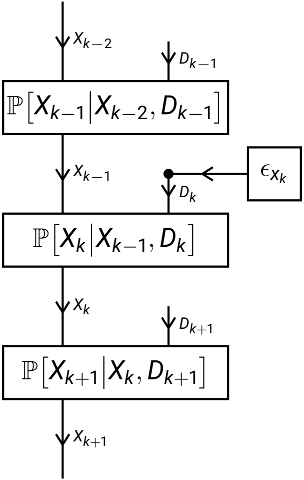
$$\mathbb{P}\left[X_k | X_{k-1}, D_k\right] = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}^{\begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}^{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}^{\begin{pmatrix} 0.50 & 0.25 & 0.25 \\ 0.25 & 0.50 & 0.25 \\ 0.25 & 0.25 & 0.50 \end{pmatrix}}}} \quad [X_k, X_{k-1}, D_k]$$

Causally augmented Markov chain



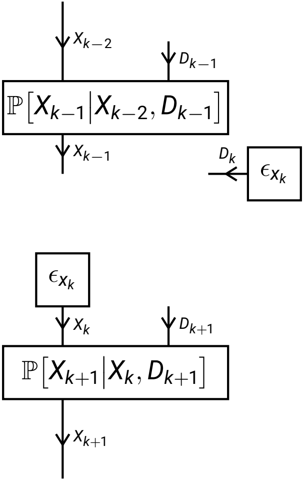


Intervening on a position



Interpretation: Restart of the random walk at time k in intervention state x_k.

Intervening on a position



Interpretation: Restart of the random walk at time k in intervention state x_k.

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Counterfactual worlds

- A particular state has been observed.
- We reason about a hypothetical world.

This appears in important situation:

- Would the accident have happened had the safety protocol been respected?
- Would the person have developed cancer had he stopped smoking earlier?

Response variables

Model each conditioned probability as a basis encoding of a function on the parent variables, the do-variable and the response variable

$$L_k$$

The response variables are shared with the counterfactual world variables.

Random walk example

We model a response variable

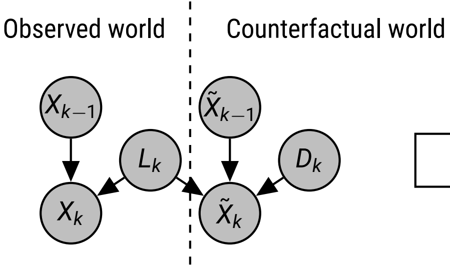
- $\ell_k = 0$: going to the left cell
- $\ell_k = 0$: staying at the current cell
- $\ell_k = 2$: going to the right cell

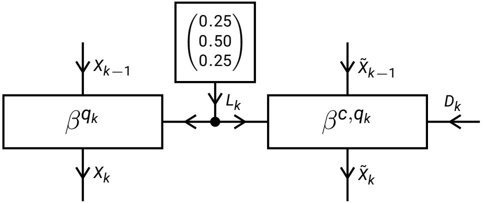
$$\mathbb{P}\left[X_k | X_{k-1}, L_k\right] = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}^{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}}} \left[X_k, X_{k-1}, L_k\right]$$

Effect in random walk: Shift of the random walk.

Counterfactual random walk

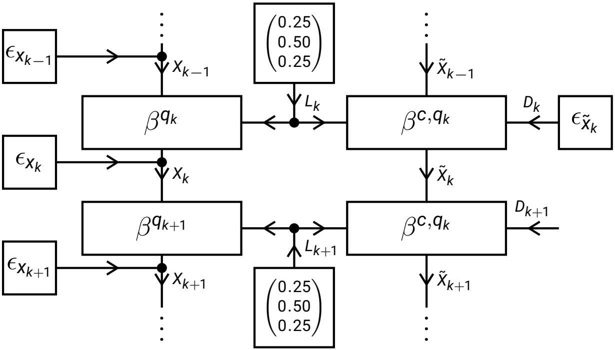
What will be the outcome of the walk, when we do an intervention on X_k?





Counterfactual query

What would be the outcome of the walk, would we have done the intervention on X_k?



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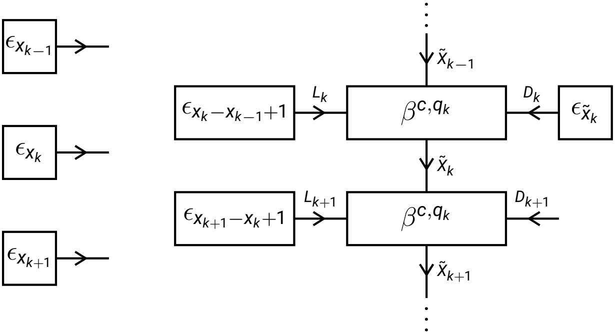

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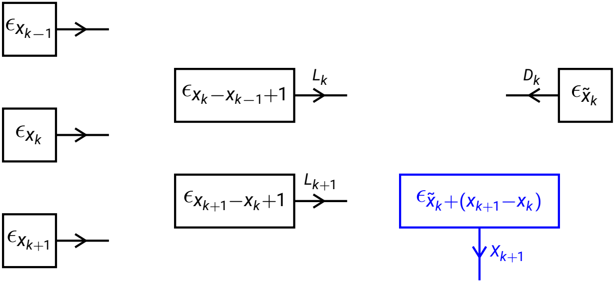

Answer: The walk would be shifted by the intervention. Different to the intervention query, we have a deterministic outcome!